QAM-I-111

Operation and Calibration
Of the Conductivity Meter

Revision 7

Major Revision Changes not indicated

Approval:

Laboratory Manager/
Laboratory Quality Assurance Officer

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Concurrence

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1.0 Applicability and Purpose

i. This procedure applies to the operation and calibration of the YSI Model 3200 conductivity meter. This procedure is performed prior to any analysis using this meter. By performing the calibration procedure, the technician reduces anomalies due to instrument sensitivity fluctuations. The operation of this instrument allows the analyst to determine the level of specific conductance of water samples received by the TIAER chemistry laboratory.

2.0 Definitions

i. Conductivity- A measure of the ability of an aqueous solution to carry an electric current. This ability depends on the presence and concentration of ionic species and the temperature of the measurement. Conductivity is customarily reported as Specific Conductance at 25ºC in micromhos per centimeter squared (µmho/cm^2) or microSiemens per centimeter (µS/cm), where 1µS/cm = 1 µmho/cm^2.

ii. Automatic temperature compensation- internal temperature sensors in the conductivity cell are read by the meter and the analysis reading is adjusted for any difference between the sample temperature and standard operating conditions. This is normally turned off during operation and compensation is mathematically performed (See SOP-C-113).

iii. Conductivity cell- probe attached to conductivity meter containing platinum electrodes used by the instrument to measure ability of the sample to carry an electrical current.

iv. Standard QA/QC definitions may be found in QAM-Q-101, "Laboratory Quality Control".

3.0 Equipment and Reagents

i. Equipment
   a. YSI Model 3200 Conductivity Meter or equivalent
   b. YSI Model 3200 Flow-Thru Conductivity Cell or equivalent with associated tubing and/or funnel connected
   c. YSI Model 3256 Immersible (dip) cell or equivalent

d. Class A volumetric flasks

ii. Reagents

a. Deionized water (DI)- water that has passed through ion exchange resin and meets Type II criteria (specific conductance < 1.0 μS/cm).

b. Standards

i. 0.01M KCl Standard Reference Solution. Dry several grams ACS grade or better KCl at 104°C overnight and cool in a desiccator. Dissolve 0.7456 grams KCl in deionized water and dilute to 1000 ml in a class A volumetric flask. This solution has a conductivity of 1412 μS/cm at 25ºC.

4.0 <u>Procedure</u>

i. Instrument Set Up

a. Ensure that conductivity cell is aligned with arrow for proper connection.

b. When turned on, meter is in Conductivity mode (uS/cm displayed). If not, then press the CELL key to enter the cell menu. Then press the CONFIGURE key to enter the configuration menu. Press the UP or DOWN keys until the correct mode is displayed on the meter screen.

c. When done, press the MODE button twice to return to the Operation menu.

ii. Calibration

a. Initial determination of cell correction factor (or if the cell is dropped or significantly altered)

i. Slowly pour some of the 0.01 M KCl conductivity solution (about 100 mL for initial flush, then about 100 mL more) through the flowcell while avoiding introduction of bubbles into the stream. For the dip cell, insert it into a container of the solution to rinse the cell, then refill the container and place the cell back into the fresh solution. Record the reading and temperature displayed on the meter in SOP-C-113-1, "Specific Conductance Logbook" or E-log.

ii. Perform four determinations of the calibration using an initial correction factor of 1.0.

iii. Determine the cell correction factor by dividing the standard concentration (1412 µmho/cm^2) by the mean of the four measurements.
iv. Record the determined constant in the log and on the cell itself beside the cell identification.
v. Determine the constant periodically or whenever significant changes occur to the cell.
vi. Factory determined constants are acceptable.

b. Measure samples or standards as described in 5.2.1.1.
c. Correct the reading to the conductivity at 25°C, using the following formula:

$$\text{Specific Cond. Corrected} = \frac{\text{Specific Cond. measured}}{1 + 0.0191\,(T - 25)}$$

d. If the corrected specific conductance is 1412 +/- 71 (1341-1483) the instrument is calibrated within specifications and may be used.
e. If the cell has a correction factor, multiply or divide by this factor to obtain the final result measurement.
f. Record the measurement in the Specific Conductance Logbook (E-log).
g. Rinse cell thoroughly with DI water.

iii. Sample Measurement

a. Make sure meter is setup to read conductivity in the correct units of µS/cm and that automatic temperature compensation is off.
b. Pour sample slowly through flowcell taking care not to introduce bubbles into the stream, allowing time for the reading to stabilize and for the cell to rinse with sample (about 100 mL). For the dip cell, rinse with the sample in a container, then refill the container. Place the dip cell in the container.
c. Read the value displayed on the meter screen and record conductivity in µS/cm (µmho/cm^2) and temperature in ºC. Make corrections for temperature and cell constant as appropriate.
d. Rinse the cell with DI before analyzing the next sample.

e. Refer to SOP-C-113, “Determination of Specific Conductance” for measurement and conversion of conductivity in water.

5.0 **Quality Control and Safety Aspects**

i. All aspects of this procedure comply with QAM-Q-101, “Laboratory Quality Control”, and QAM-S-101, “Laboratory Safety”.
ii. The cell is cleaned and rinsed thoroughly with DI water prior to and following use.
iii. The accuracy and precision of sample measurements are dependent upon a stable reading. Introduction of bubbles into the cell flow and major changes in flow velocity are avoided.
iv. Visually inspect flow path to ensure there are no obstructions, which may alter sample/standard solution flow.
v. Record all standards in the Standards Logbook (E-log).
vi. The analyst consults the MSDS files if he/she has any question as to the safe handling of any reagent required by this procedure for analysis.
vii. Safety glasses, at a minimum, are worn at all times when performing this procedure.

6.0 **References**

i. YSI Model 3200 Operations Manual, Revision D, YSI Incorporated, July, 1998, Yellow Springs, Ohio, ITEM# 003224.
ii. Standard Methods for the Examination of Water and Wastewater, latest online edition, Washington D.C., Method 2510 B.
iii. The National Environmental Laboratory Accreditation Conference Institute (TNI) standard, 2009.

7.0 **Attachments**

None